

A Ring Topology Approach: Efficient Communication in Wireless Body Area Networks (WBANs)

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Abstract—The ongoing developments in Beyond 5G (B5G) and Sixth Generation (6G) wireless systems are shifting the human factor increasingly back into focus, with a particular emphasis on Wireless Body Area Networks (WBANs). Therefore, in this work, a new Printed Circuit Board (PCB) around the ESP32-S3 microcontroller as well as an innovative protocol based on ESP-NOW optimized for reliability and low latency are proposed. The developed PCB is designed as a modular, flexible sensor carrier for biometric applications. In addition to the multi-hop and token-based approach to reduce the computing load and the communication volume, the core of the development is the implementation of a ring network topology, as well as the increase in efficiency and reduction of energy consumption. The results of the work prove a reduced overall latency of approximately 50 % in combination with a 36 % lower power consumption compared to traditional star network topologies on the same hardware. Therefore the protocol not only shows an increase in efficiency for state-of-the-art systems but also proposes a flexible sensor carrier board for biometric data acquisition while reducing the overall amount of communication needed.

Index Terms—Beyond 5G, 6G, Wireless Body Area Networks (WBANs), ESP-NOW, Wireless Sensor Networks (WSNs), multi-hop WBAN

I. INTRODUCTION

The field of wireless communication systems has undergone significant development over the last few decades, whereby each generation has been accompanied by advances in speed, capacity, and areas of application. Especially with regard to the current development in Beyond 5G (B5G) and Sixth Generation (6G) wireless systems, this creates social and technological requirements in terms of performance, scalability, and efficiency, but it also opens up opportunities and additional areas of application, for example in the health and lifestyle sector [1]. The holistic integration of different technologies—such as Wireless Optical Communication (WOC), the use of frequencies in the (sub)terahertz (THz) range or Integrated Sensing and Communication (ISAC)—, and the goal of Ultra-Reliable Low-Latency Communication (URLLC) will enable applications such as Holographic-Type Communication (HTC), remote surgery, real-time health monitoring, and personalized medicine [2].

In this domain, technologies such as Wireless Sensor Networks (WSNs) and Wireless Body Area Networks (WBANs) have been established, former implemented in the corresponding standards *Institute of Electrical and Electronics Engineers (IEEE) 802.15.6 Wireless Body Area Network* [3] and *IEEE Std. 11073-20701-2012 Service-Oriented Medical Device Exchange Architecture and Protocol Binding* [4]. With the ability to continuous patient monitoring, data collection, processing, and transmission, as well as improved security features

through the use of biometric factors, these network types offer enormous potential for improved patient care [5]. However, their full potential has yet to be realized, primarily due to challenges in achieving efficient, low-latency communication while ensuring energy conservation. Current implementations, most of which rely on centralized or star network topologies, inherit limitations regarding scalability, flexibility, and energy efficiency. Therefore, it is important to design a system not only aligned with the 6G vision of decentralized intelligence but also considering the specific nuances of healthcare applications, especially when dealing with critical biometric data [6]. To cover this, a novel ESP-NOW-based low-latency protocol designed explicitly for WBAN applications and implemented on an ESP32-centric Printed Circuit Board (PCB) is introduced in this work. Through leveraging a token-ring mechanism, the proposed system offers enhanced communication efficiency, dynamic adaptability, and optimal energy consumption.

Particularly considering applications in ambulant pre- and post-clinical environments, the smart hospital metaverse, and telemedical aspects, the proposal is highly attractive for industrial and clinical partners. Given the current and future challenges, sustainability, energy efficiency coupled with security and data protection compliance are essential factors for a successful roll-out [7]. Above all, the topic of security is incredibly important but is not the focus of this work. However, corresponding work such as [7]–[10] has been taken into account.

Although the methodology of the presented work shows characteristics of a multi-hop system and, as such, could potentially extend the transmission range of the overall system, the focus of this study is on its application as WBAN. Consequently, this methodology will be referred to as “token-based” in the subsequent text.

To showcase the potential of the work in the context of the 6G development, Section II gives an overview of state-of-the-art WBAN approaches. Furthermore, Section III introduces the hardware configuration and the design of the ESP-based sensor carrier. The overall communication protocol and software implementation are described in Section IV, whereas in Section V the performance results and a comparison of the token-based approach with the traditional star-like system are conducted. Finally, Section VI summarises the work.

II. WIRELESS BODY AREAS NETWORKS: AN OVERVIEW

Alongside the traditional communication standards used in the Internet of Things (IoT) context, such as IEEE 802.11 Wireless Local Area Networks (WLANs), Long Range Wide

Area Networks (LoRaWANs) and Fifth Generation (5G), the IEEE 802.15.6 WBANs standard has been established, particularly in the healthcare domain and in networks close to the human body [3][11].

A comprehensive introduction and study on Media Access Control (MAC) protocols customized for WBANs is given by *Movassaghi et al.* They are emphasizing the need to optimize energy efficiency in addition to reliable communication, which is essential for devices in close proximity to or within the human body [12]. Besides, *Yang, Wang & Zhaolin* [13] present a MAC protocol for WBANs combining different access methods. Their approach indicates promising results in terms of energy saving, -which is still a key criterion for practical WBAN implementations-. Furthermore, *Ghaffari & Ghaffari*, highlight the challenges and potential solutions of Energy Efficient Routing Protocolss (EERPs) for WSNs, underlyning the importance of efficient routing in WBANs [14].

In addition, *Cornet et al.* provide a holistic view of WBANs, particularly in the area of mobile healthcare. They discuss the unique challenges posed by WBANs communication, particularly emphasizing the nuances of 60 GHz millimeter wave (mmWave) technology. Furthermore, the work addresses the integration of WBANs into the evolving landscape of 5G networks and provides an outlook on the integration of these networks into the expected 6G framework [1].

With regard to token-based protocols, relevant contributions have been made, for instance, by *Wei et al.*, proposing a modified version of Wireless Token Ring Protocol (WTRP), adapted to sensor networks and focusing on self-organization, fairness in node access and Quality of Service (QoS) [15]. In contrast, *Zheng, Zhang & Zhang* describe their concept of a receiver token ring to avoid collisions in densely populated wireless networks. Their approach leveraged the capabilities of virtual token ring networks to minimize collisions, especially in areas with a high concentration of nodes [16].

To ensure seamless communication in such networks, *Eridani et al.* compare the strengths of the well-known communication protocols, Wi-Fi, Bluetooth, and ESP-NOW, recognizing ESP-NOW for its distinctive performance metrics [17]. In addition, *Labib et al* use the ESP-NOW protocol to build an adaptive spider mesh topology. Their approach provides a larger communication range and showed a 15-fold improvement in distance over Bluetooth Low Energy (BLE), but at twice the power consumption [18]. On a similar note, *Maier et al* conducted a comparative analysis of the ESP32 Microcontroller Unit (MCU) module and highlighted its benefits for IoT applications. This insight provides a deeper understanding of why the ESP ecosystem is suitable for WBAN deployments [19].

Building on the aforementioned fundamental studies, this work introduces a novel token system protocol for WBANs. This approach is mainly characterized by its validation on real hardware and aims to combine the theoretical robustness of previous work with tangible, empirical verification.

III. SYSTEM DESIGN AND METHODS

This section outlines the development of the custom sensor-carrier PCB, designed around the ESP32-S3 chip. It discusses the specifics of the hardware implementation, detailing the components. Therefore, Fig. 1 provides an overview of the PCB-Design, whereas Fig. 1 a) indicates the sensor carrier with an attached sensor board, and b) gives an overview of the used hardware components.

A. Dual-Core Architecture and Multitasking

The ESP-NOW-based communication system is built on an ESP32-S3 MCU, taking advantage of dual-core functionality: one core manages the acquisition and processing of sensor data, while the other core manages the transmission of data via the ESP-NOW protocol. This dual-core architecture, complemented by the use of FreeRTOS, ensures effective multitasking, a crucial element in maintaining low-power operation and wireless communication. The programming interface is facilitated by the CP2102 USB-to-UART bridge IC, simplifying tasks such as firmware programming, debugging, and serial communication.

B. Power Supply and Management

To ensure the longevity of the 3.7V LiPo battery in the design, a voltage divider circuit and a special Undervoltage Protection Integrated Circuit (IC) are incorporated. The PCB's four-layer design, including ground layers, is displayed in Fig. 1, which enhances electrical performance. Stability in power supply to the MCU and related components is maintained by a 3.3V Low Drop Out (LDO) regulator. The system's status is indicated through three LEDs, showcasing Power Good, Busy/Alive, and Software Error.

C. External Components and Sensor Shields

To extend the system's capabilities, external components such as 32Mbit Flash and 64Bit external RAM are soldered to the underside of the 4-layer board. This additional hardware increases storage capacity and memory, supporting more extensive data handling and processing capabilities. Sensor data acquisition is facilitated by customized sensor shields that can be seamlessly attached to the system via the small 1.27mm pitch headers. These shields, specifically designed for different sensing tasks, offer a modular approach to integration, allowing the sensor system to be easily expanded and customized. This adaptability positions the system as an ideal solution for a wide range of IoT applications.

IV. COMMUNICATION PROTOCOL - SOFTWARE IMPLEMENTATION

The entire communication protocol, developed based on ESP-NOW and written in C programming language, is described here. First, the token-based ring structure is explained,

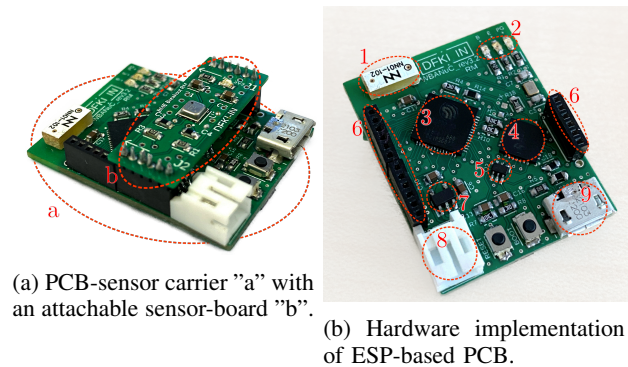


Fig. 1: PCB-Overview (a), and Hardware design (b) of Sensor node, 1: Antenna, 2: three control LEDs, 3: ESP32-S3-MCU, 4: USB-Bridge, 5: Bootstrapping-mechanism, 6: SensorLink Interfaces, 7: Power management, 8: battery connector, 9: Programming Interface.

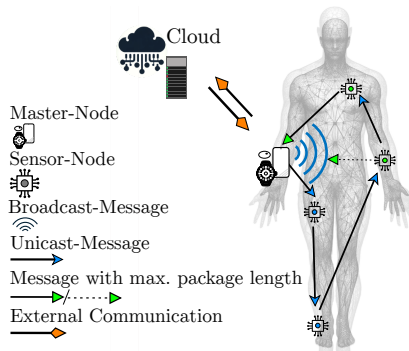


Fig. 2: Schematic overview of WBAN token-based ring communication approach, illustrating the communication path across different sensor nodes at a body. A blue arrow indicates the aggregation of sensor data (Unicast-Message), while a green arrow highlights a full package ("Full"-Master-Message). Communication with a cloud or server application (yellow arrow) for further analysis takes place from the master node.

followed by explaining the transmission including defined headers for data assembling of the different sensor nodes on the server side¹.

A basic overview is provided in Figure 2, presenting a schematic representation of the communication flow within the network. The transmission cycle is depicted, starting from the master node, which runs through five sensor nodes and returns to the master node. This master node can also forward the collected data to a cloud server for further processing and analysis.

A. Ring Topology Communication

The three different phases: i) initialization of the system, ii) pathfinding of the sensors, and iii) the communication routine itself, are specified in Figure 3. Together, these enable robust and efficient data exchange over the network. In more detail, these 3 phases consist of:

a) *Initialization*: The master unit (M) commences the system by broadcasting a signal awaking all nearby actively used sensor devices ($S_0, S_1, \dots, S_n$) from standby mode, prompting them to transmit their information to the master. Each slave, in turn, encapsulates its data (MAC address, data type, typical data size, etc.) and sends it to the master following a random delay between 0 and 1 second to prevent signal interference. The master sequentially stores all incoming data in a buffer, arranged by the order of arrival. This buffer is then broadcasted to all slaves, ensuring they are aware of the MAC address assignments corresponding to specific slave indices, which is crucial for the operation of the subsequent token train. In the communication routine's cycle, as shown in Fig. 3, within the order of arrival of the sensor nodes, the master node generates the roadmap, dispatched as a broadcast in each cycle during the communication routine (c). This roadmap, a 16-bit sequence, serves as a guide for sensor node participation in the forthcoming cycle: a '0' bit indicates a node's exclusion, whereas a '1' denotes its inclusion. This binary representation efficiently dictates the active participation of each sensor node, streamlining the communication process within the network.

¹ Please note: As this exceeds the scope of this work, and as the focus is on the communication protocol, the necessary security aspects and the encryption within the sensor system are not dealt with here!

b) *Pathfinding*: The master emits a broadcast to initiate the "First Call." Subsequently, the slaves transmit messages in sequence to the next slave with a higher index. The process starts with Slave 0, triggered by the master's broadcast signal. This method determines the typical cycle time for the current configuration and confirms that the registration was successful and that all slaves are accessible. The broadcast initiating the "First Call" also conveys a timestamp from the master, providing each slave with a time relative to the master. Thereafter, the frequencies of the slaves and the typical data sizes are sorted, laying the groundwork for calculating bus schedules.

c) *Communication routine*: A cycle, defined as one roundtrip from the master over the sensor nodes back to the master, initiates the formulation of the current bus schedule. A 16-bit value is generated, where each bit indicates whether a slave is a station on this current plan. This plan or 'roadmap' is then broadcast to every slave. Each slave interprets this schedule and sets timers based on their actual position in the roadmap to prepare for potential bus loss. The first slave, acknowledging its position, immediately sends its bus message via unicast to the next station. The receiving slave appends its package and forwards it to the next station, continuing until the last slave, which sends the package directly to the master.

B. Special Situations in WBAN System

In addition to standard operations, the WBAN system handles several special cases:

- i) When a slave node's data exceeds bus capacity, it sends a minimal token message to the next station for bus continuity, followed by the incomplete bus to the master.
- ii) If a slave's data surpasses transmission limits, it is sent in maximum-sized packets to the master until the final part, which either continues as in scenario i) or as a regular bus message.
- iii) The master node synchronizes to send the previous cycle's message to an external node while awaiting the next bus, ensuring no delay in the system.
- iv) If the bus schedule is entirely zeros, the master delays the system for a cycle time determined in the Pathfinding step.
- v) After a set number of cycles, the system either reverts to Initialization or proceeds with a new Pathfinding routine.

A maximum of 16 nodes can be managed by one master node. For additional sensors, a secondary master node is implemented, communicating directly with the primary master node.

C. Communication Setup - Added Header Architecture

For the overall communication protocol, new headers had to be defined, to provide a sufficient exchange of information between sensor nodes and the master node. A short overview of the headers and their purpose is given in the following.

1) *General Message Header*: The general message header is included in every message or piece of message between sensor nodes inside the token-based protocol. It consists of:

- A 4-byte header that is used for each message that is to be sent in the communication system.
- The msg_type field always contains an indicator of the context of the current message, i.e. error types.
- The slave-index field is used to identify the sender of the current message. This is important for quickly checking that the token train is following the correct path.

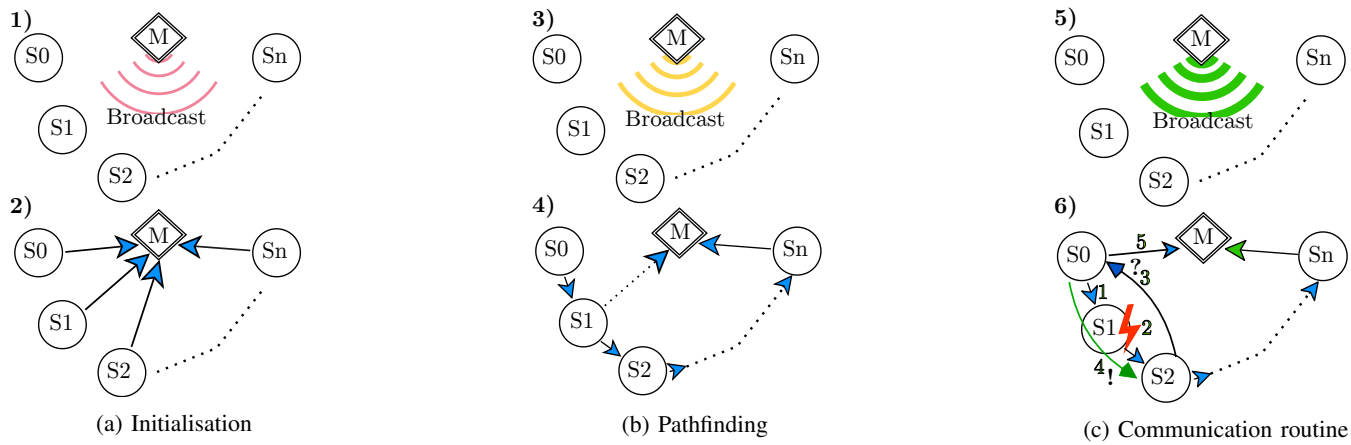


Fig. 3: Shown is the overall protocol of the token-based communication approach which is delineated into three distinct phases: Initialization (a), Pathfinding (b), and Communication routine (c). During the Initialization phase, the master unit emits a broadcast signal that is detected by authenticated sensor nodes (1). These nodes, in turn, reply to the master with essential information unique to each (2). In the Pathfinding phase, the master unit dispatches a 'First Call' broadcast (3) to calculate the round-trip time and confirm the successful enlistment of the sensor nodes (4). Finally, the communication routine phase involves the transmission of data throughout the network. This phase begins with the master unit sending a roadmap (5) informing the sensor nodes of the next nodes to communicate with (6). In the event of delays 1), 2), an error message is triggered by the first affected slave 3) and directed to the slave positioned two nodes prior. This process is repeated iteratively until the problem is resolved 4). The first responsive slave also involves the master in this error management process, who then takes the necessary steps 5).

- The correctness of each message is checked with a Cyclic Redundancy Check (CRC). The 3rd and 4th bytes in the header are used to store the required CRC value.

2) **Tokentrain Header:** The tokentrain header is a 7-byte additional header used by the slaves when encapsulating their data for the current iteration of the token train. It is mainly used to classify the data when it is evaluated externally, but it is also used to verify that the system is functioning correctly. It consists of:

- The 1-byte msg-length field holds the length of the sending slave's data, length of the tokentrain-header included. This is used to parse the tokentrain-package while still allowing variable send-lengths successfully.
- 4 bytes of the header are used to hold a timestamp relative to the master time. This is mainly used for data evaluation and system performance analysis.
- The 1 Byte Slavetype field is also used for classification by indicating the type of the sending module, for instance, medical-related sensors such as temperature-sensor, EEG, EMG, ECG.
- The last field in the token train header is the Slaveindex. This is the same value as in the general header and is used for quick verification.

When the tokentrain travels, the general message header of the previous slave is always discarded and renewed, but the tokentrain header is retained, shown in Fig. 4, 'Message setup'.

The difference between the headers is that the General Message header is used for direct communication within the system, whereas the Tokentrain header is used more as a tool for external classification and evaluation of the data.

General message header

msg_type	Slave-index	CRC-Value
8 Bit	8 Bit	16 Bit

Tokentrain header

msg-length	Timestamp	Slavetype	Slave-Index
8 Bit	32 Bit	8 Bit	8 Bit

Message setup

general message header	sensor node 1	...	sensor node n
	tokentrain header 0		tokentrain header n
	data 0		data n

Fig. 4: Header structure of token-based communication approach - from top to bottom it shows the general message header, the tokentrain header, and the overall message setup during a communication cycle.

D. Error Management

An error management process is implemented upon system start, focusing on potential disruptions post-bus schedule broadcast. Each slave node activates a timer, crucial for synchronization with the bus. If the bus fails to arrive in time, the node launches an error protocol, alerting an earlier node in the schedule. This step helps identify issues like lost messages or crashes. A broadcast message from the receiving node prompts the master node and originating nodes to determine further action. If unresponsive, the process iteratively moves back in the schedule, involving the master node for potential isolation of malfunctioning sensor nodes, thus ensuring system stability and communication flow.

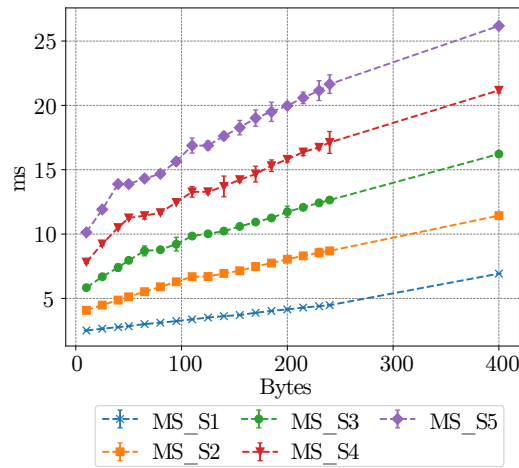


Fig. 5: transmission rate of token-based system from one Node (N1) up to five Nodes (N5) with the data sizes per sensor node from 10 bytes up to 240 bytes in steps of 15 bytes and 400 bytes as it exceeds the maximal package length of 250 bytes.

V. EVALUATION AND RESULTS

The token-based communication system is compared against a traditional star-configured ESP-NOW system. Two primary metrics are analyzed for this comparison: i) Power consumption and ii) Latency/ Transmission rate of the system, varying the number of sensor nodes and data length for each node.

One master unit communicates with up to 5 independent sensor nodes in a hardware setup, whereas the master unit itself sends the transmitted data to an observer unit which is connected to a PC, storing collected data. Dummy data is sent through the network to test the performance in terms of the transmission rate of the overall system with different data sizes per sensor node and measurement series. For comparison, a state-of-the-art ESP-NOW approach is created, and the master unit communicates with each sensor node, one after the other in a star schema. The tests were conducted in a

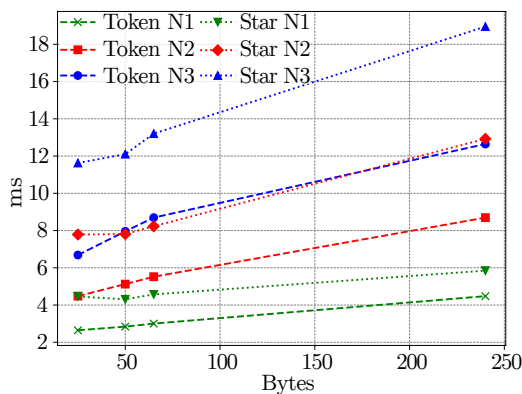


Fig. 6: Comparison of token-based (*Token*) and star topology (*Star*) system regarding transmission rate. The Graph shows the needed latency for the system to receive all data of all sensor nodes regarding the number of used nodes (*N1-N3*) and data size of 65 bytes per sensor node. A smaller number means a better latency time and a faster transmission rate. It shows that the token-based approach outperforms the star system within all approaches.

controlled environment replicating real-world conditions, by 23 degrees Celsius. Measures were implemented to prevent interference from nearby Wi-Fi networks. The data for both the token train or token-based approach and the star system were measured under identical conditions, ensuring consistency and comparability of the experimental results.

A. Transmission Rate and Latency

The required transmission rate and latency of the token-based approach are tested from only one sensor node, communicating with the master unit up to 5 sensor nodes, communicating within a cycle between sensor nodes and the master unit. Therefore, the transmission rate of a whole cycle with data sizes per sensor node from 10 bytes up to 240 bytes with 15 bytes-steps is collected. Each cycle includes thereby the communication setup of the master unit and the data aggregation of all sensor nodes back to the master unit, transmitting the data back to the PC. Fig. 5 shows the transmission rate of the token-based system from one sensor node up to five sensor nodes. It describes the required transmission rate for a whole cycle applied against the transmitted bytes. Therefore, for every pair of sensor nodes and byte rate 10 minutes of data and the corresponding transmission rate are collected, consisting of approximately 200 data points per measurement. For the value in the transmission rate diagram, the mean value of the rate over the 10 minute data acquisition is taken. Error bars show the standard deviation of the transmission speed during data collection.

Based on the values depicted, it can be deduced that the transmission delay of each respective byte value from a system of one more or one less sensor node to each other is equidistant by ± 0.4 ms. Till an overall byte rate of 240 bytes a slight logarithmic progression can be seen, after that the curve will get a linear character. If the data size per node exceeds the 240 bytes, each sensor node communicates back to the master node by sending the 'full package' and can only send a new header file to the next sensor node. Therefore the most efficient way to use the token-based system approach is for data sizes per node below the maximum package length.

The comparison between the token-based and the star-shaped system, with a system configuration of one sensor node (N1) to three sensor nodes (N3), is given in Fig. 6. Therefore, a constellation with a data size of (25, 50, 65, 240) per sensor node is used. It is shown that the token-based approach outperforms the state-of-the-art approaches in all configurations. Table I shows the ratio between the star and token-based approach and with which factor the comparable approach is getting outperformed. It shows, that the system best performs with low data sizes, but stabilizes with around

TABLE I: The table presents a comparison of transmission rates, highlighting the efficiency of the token-based approach against the star systems. It quantifies the performance gain of the token-based system via the number of nodes included and the corresponding data size.

Bytes	1 node	2 nodes	3 nodes
25	1.68	1.74	1.74
50	1.52	1.52	1.52
65	1.52	1.49	1.52
240	1.31	1.49	1.50

50 % better latency starting with two nodes also with the maximal data size of 240 bytes. Also, if the system only works with one node with maximal package length, the system outperforms the state-of-the-art approach, because of the more efficient transmission protocol and header structure.

B. Power Consumption

To measure the overall performance the power consumption of the token-based and star system is measured. Therefore, for three times both systems are running with a data size of 65 bytes per sensor node & cycle and 3 sensor nodes. The power consumption is measured and a similar 250 mAh battery is used for both systems. Table II gives the results of both systems. It can be seen, that the power consumption of the star system is 36 % more compared to the token-based approach, while the token-based approach has a lower latency and transmits data 52 % faster.

TABLE II: Energy consumption of star and token-based approach with 3 sensor nodes and data throughput of 65 bytes per cycle with system-dependent maximal communication speed.

System	Runtime with 260 mAh	Power consumption
Star like	2 h and 10 min	118.2 mA/s
Token based	3 h	87.0 mA/s

VI. CONCLUSION AND OUTLOOK

This work introduces an innovative ESP-NOW-based protocol, implemented on a newly developed ESP32-centric PCB. The token-ring mechanism central to this study demonstrates a novel approach to distributing computational load and communication traffic, resulting in notable energy conservation during sensor data aggregation while decreasing overall latency. The comparative analysis with traditional star systems underscores the substantial improvements achieved by this protocol. Notably, the token-based system shows a significant reduction in latency and power consumption compared to the star-configured system. It was observed that with the increase in number of sensor nodes and data length, the token-based approach consistently outperformed the traditional system in terms of latency and energy efficiency. In configurations with low data rates and multiple nodes, the token-based system exhibited approximately 50 % better latency and 36 % lower power consumption compared to the star configuration. Moreover, the system provides an effective solution for the integration of body-worn sensors to relieve communication load in environments with many (inter) connected devices such as wireless headphones and smartwatches. This approach has the ability to reduce network overload and to maintain efficiency in densely WBANs.

These advancements represent a major stride in WBAN communication, aligning with the vision of 6G research for decentralized intelligence in healthcare applications. The work, in addition to that, sets a foundation for future work to explore advanced security measures and optimization for specific healthcare scenarios, emphasizing the potential of this protocol to revolutionize healthcare communication in the 6G era.

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